

MACHINE LEARNING-INTEGRATED POWER BI DASHBOARD FOR MONITORING AND PREDICTING ON-TIME STUDENT GRADUATION

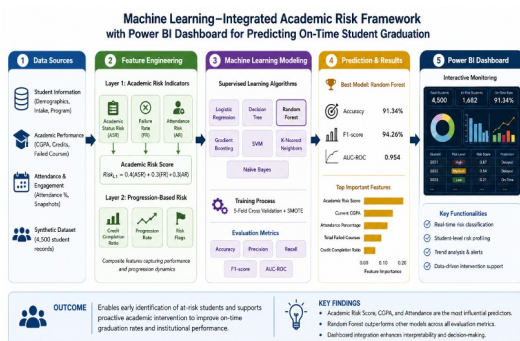
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Graphical abstract



Abstract

On-time graduation is an important performance indicator in higher education, reflecting institutional effectiveness and student success. However, existing academic monitoring systems are often limited to descriptive analytics and fragmented data, making early identification of at-risk students difficult. This study proposes a machine learning-integrated academic risk modelling framework combined with an interactive Power BI dashboard to monitor and predict student graduation outcomes. A synthetic dataset of 4500 undergraduate Computer Science students was generated to simulate realistic academic progression. The CRISP-DM methodology was adopted, and several supervised learning models were evaluated. A layered feature engineering approach was introduced to capture both academic performance and progression dynamics. After hyperparameter optimization, the tuned Random Forest model achieved the best performance with an accuracy of 91.34%, F1-score of 94.26% and AUC-ROC of 0.954. Key predictors include academic risk score, CGPA, attendance and failed courses. The model outputs were integrated into an interactive dashboard to support real-time monitoring and early identification of at-risk students. The proposed framework enables proactive academic intervention and supports data-driven decision-making in higher education.

Keywords: Student graduation prediction, machine learning, Power BI, Random Forest, predictive analytics, SMOTE.

Abstrak

Kadar tamat pengajian tepat pada masanya merupakan penunjuk prestasi penting dalam pendidikan tinggi kerana mencerminkan keberkesanan institusi dan kejayaan pelajar. Walau bagaimanapun, sistem pemantauan akademik sedia ada masih terhad kepada analitik deskriptif dan data yang terpisah, sekali gus menyukarkan pengenalpastian awal pelajar berisiko. Kajian ini mencadangkan satu rangka kerja pemodelan risiko akademik berasaskan pembelajaran mesin yang digabungkan dengan papan muka interaktif Power BI untuk memantau dan meramal hasil tamat pengajian pelajar. Satu set data sintetik yang mengandungi 4500 rekod pelajar Sarjana Muda Sains Komputer telah dijana bagi mensimulasikan corak perkembangan akademik yang realistik. Metodologi CRISP-DM digunakan dan beberapa model pembelajaran mesin terselia telah dinilai. Pendekatan kejuruteraan ciri berlapis diperkenalkan bagi menangkap kedua-dua aspek prestasi akademik dan dinamik perkembangan pelajar. Keputusan menunjukkan bahawa model Random Forest mencapai prestasi terbaik dengan ketepatan 91.34%, skor F1 sebanyak 94.26% dan AUC-ROC sebanyak 0.954. Faktor utama yang mempengaruhi ramalan termasuk skor risiko akademik, CGPA, kehadiran dan bilangan kursus gagal. Hasil ramalan telah diintegrasikan ke dalam papan muka interaktif bagi menyokong pemantauan masa nyata dan pengenalpastian awal pelajar berisiko. Rangka kerja yang dicadangkan ini membolehkan intervensi akademik secara proaktif serta menyokong pembuatan keputusan berasaskan data dalam pendidikan tinggi.

Kata kunci: Ramalan tamat pengajian pelajar, pembelajaran mesin, Power BI,

1.0 INTRODUCTION

On-time graduation is widely recognised as a key performance indicator in higher education, reflecting both institutional effectiveness and student academic success. With the increasing adoption of learning management systems (LMS) and student information systems (SIS), universities now collect large volumes of academic data that can be leveraged to analyse student progression and performance patterns [1, 2]. The emergence of learning analytics has further enabled institutions to extract actionable insights to improve student outcomes and institutional decision-making [3].

Machine learning (ML) techniques have demonstrated strong potential in modelling complex relationships within educational datasets and predicting outcomes such as academic performance, dropout risk, and graduation timelines [4, 5, 6]. In particular, supervised learning models such as Random Forest, Support Vector Machines (SVM), and Gradient Boosting are widely applied due to their ability to capture nonlinear relationships and handle complex features [7].

Despite these advancements, universities still face challenges in effectively monitoring student progress. Academic data is often fragmented across multiple systems, making it difficult for academic staff to obtain a comprehensive view of a student's academic journey [8]. Furthermore, most institutional dashboards primarily provide descriptive analytics and lack predictive capabilities, resulting in delayed identification of at-risk students and reactive intervention strategies [9, 10].

Business intelligence tools such as Microsoft Power BI have been increasingly adopted to visualise institutional data and support decision-making processes [11, 12]. However, these tools are typically limited to descriptive reporting and are not fully integrated with predictive analytics models.

The main contributions of this study are as follows:

1. A layered academic risk modelling framework incorporating performance and progression indicators;
2. A comparative evaluation of machine learning models for graduation prediction;
3. Integration of predictive analytics into an interactive Power BI dashboard;
4. A practical decision-support system for early identification of at-risk students.

Unlike previous studies that focus primarily on standalone predictive models or descriptive dashboards, this study integrates layered academic risk modelling with interactive business intelligence visualization within a unified predictive monitoring framework.

Therefore, this study proposes a machine learning-integrated Power BI dashboard that combines predictive modelling with interactive data visualization. This integration aims to enhance early identification of at-risk

students, support proactive academic interventions, and improve data-driven decision-making in higher education.

2.0 LITERATURE REVIEW

Learning analytics has become a central component in improving student performance and institutional effectiveness. Traditional academic dashboards mainly provide descriptive insights and require manual interpretation to identify at-risk students [1], [3].

Machine learning approaches have been widely applied in predicting student performance and graduation outcomes. Algorithms such as Random Forest, Logistic Regression, Support Vector Machines, and Neural Networks have demonstrated strong predictive capabilities in educational data mining [4, 5, 6]. Academic indicators such as CGPA, attendance, and course failures are consistently identified as strong predictors of student success [4], [13].

Ensemble learning methods such as Random Forest and Gradient Boosting are particularly effective due to their robustness and ability to capture complex feature interactions [7], [14].

To address class imbalance in educational datasets, techniques such as Synthetic Minority Oversampling Technique (SMOTE) are widely used [15, 16]. In addition, explainable artificial intelligence (XAI) approaches such as SHAP have been applied to improve model interpretability and provide insights into academic risk factors [15].

Beyond academic factors, recent studies highlight the influence of non-academic variables such as student engagement, socio-economic background, and behavioural patterns in predicting student outcomes [2], [17].

Business intelligence tools such as Power BI are widely used for visualising academic data and supporting decision-making [11, 12]. However, most implementations remain descriptive and lack predictive integration.

Recent research emphasises the importance of integrating predictive analytics with visualization tools to enable real-time monitoring and proactive intervention [9], [18]. Despite these advancements, there remains a research gap in developing structured frameworks that integrate predictive modelling, interpretable risk analysis, and interactive visualization in a unified system.

3.0 METHODOLOGY

This study adopts the CRISP-DM framework to guide the data mining process, including data understanding, data preparation, modelling, evaluation, and deployment phases (Figure 1).

A synthetic dataset consisting of 4500 undergraduate Computer Science student records from intake cohorts between 2021 and 2025 was generated based on institutional academic structures [19, 20]. The dataset

includes demographic, academic performance, and progression-related attributes as shown in Table 1. The synthetic dataset distributions were designed based on institutional academic regulations and typical student progression patterns observed in undergraduate programmes.

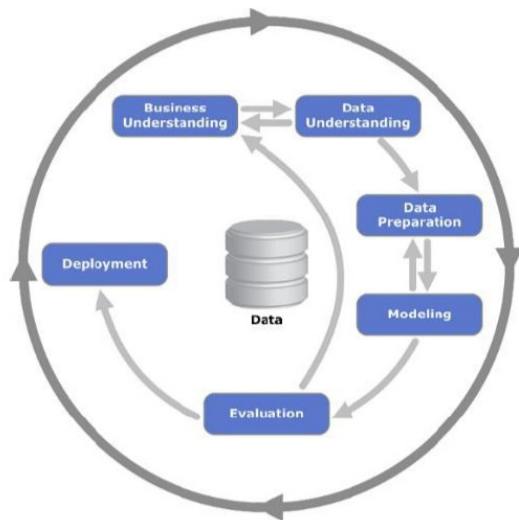


Figure 1 CRISP-DM Methodology

Table 1 Data Dictionary of the Generated Student Dataset

No.	Variable Name	Type	Category	Description
1	Student_ID	Numerical	Student Information	Unique identifier assigned to each student record
2	Intake_Session	Categorical	Intake Information	The intake period when the student enrolled
3	Gender	Categorical	Demographics	Gender of the student
4	Race	Categorical	Demographics	Ethnicity of the student
5	Nationality	Categorical	Demographics	Country of origin of the student
6	Attendance_Percentage	Numerical	Academic Performance	Overall attendance percentage of the student
7	Current_Year_of_Study	Numerical	Academic Progress	Current year of study in programme
8	Current_Semester_of_Study	Numerical	Academic Progress	Current semester of study
9	Current_CGPA	Numerical	Academic Performance	Current cumulative grade point average
10	Current_Total_Credit_Hours_Earned	Numerical	Academic Progress	Total credit hours successfully completed
11	Total_Failed_Courses	Numerical	Academic Performance	Total number of failed courses

No.	Variable Name	Type	Category	Description
12	Snap3_Attendance	Numerical	Snapshot Variables	Attendance percentage recorded at Semester 3 snapshot
13	Snap3_CGPA	Numerical	Snapshot Variables	CGPA recorded at Semester 3 snapshot
14	Snap3_Credits	Numerical	Snapshot Variables	Total credits accumulated by Semester 3
15	Snap3_Failed	Numerical	Snapshot Variables	Number of failed courses recorded by Semester 3
16	Snap4_Attendance	Numerical	Snapshot Variables	Attendance percentage recorded at Semester 4 snapshot
17	Snap4_CGPA	Numerical	Snapshot Variables	CGPA recorded at Semester 4 snapshot
18	Snap4_Credits	Numerical	Snapshot Variables	Total credits accumulated by Semester 4
19	Snap4_Failed	Numerical	Snapshot Variables	Number of failed courses recorded at Semester 4
20	Internship_Completed (Yes/No)	Binary	Academic Progress	Indicates whether the student has completed the internship requirement
21	Graduation_Session	Categorical	Graduation Information	Session in which the student graduates
22	Student_Status	Categorical	Graduation Information	Current academic status
23	Outcome	Binary	Target Variable	Final classification whether the student graduates on time or delayed
24	Final_Semesters_Taken	Numerical	Graduation Information	Total number of semesters taken to complete the programme

In addition, data preprocessing was performed to ensure data quality through cleaning, duplicate removal and range validation. Feature engineering was also applied to derive additional variables that better represent students' academic progression. The engineered features were categorized into two groups which are engineered Layer 1 and Layer 2 (see Table 2). Layer 1 features focus primarily on academic performance indicators and student risk classification while Layer 2 features focus on credit

progression and academic progress relative to the programme structure.

Table 2 Engineered Layer 1 and Layer 2 Features

Layer	Example Features
Layer 1	Academic Risk Score, Attendance Risk
Layer 2	Credit Completion Ratio, Progress Delay Index

Seven supervised machine learning algorithms were implemented and compared including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine (SVM), K-Nearest Neighbours (KNN) and Naïve Bayes. The dataset was divided into training (80%) and testing (20%) subsets using stratified sampling and Synthetic Minority Oversampling Technique (SMOTE) was applied to address class imbalance between on-time and delayed graduation cases. Feature scaling using StandardScaler was also applied and the best performing model was further optimized using GridSearchCV with 5-fold cross-validation. These models were selected due to their proven effectiveness in handling tabular educational datasets and their ability to capture both linear and nonlinear relationships.

Model performance was evaluated using accuracy, precision, recall, F1-score and AUC-ROC metrics. The best model was then integrated into the proposed system where prediction results were visualized through an interactive Power BI dashboard. The dashboard displays student demographics, academic indicators, and predicted graduation risk levels using interactive charts and filters. This enables academic staff and advisors to monitor student progress and identify at-risk students for early intervention.

The proposed system follows a linear data pipeline as shown in Figure 2 that transforms raw student data into predictive insights. The process begins with a synthetic historical student dataset which undergoes data preprocessing for data analysis. A correlation test is then conducted to identify relevant variables followed by feature engineering and label creation to construct key features and define the target variable. The processed dataset is applied to multiple machine learning algorithms to develop prediction models and model performance is evaluated to determine their effectiveness. The final prediction results are exported as a CSV file and visualized in an interactive Power BI dashboard for monitoring and interpretation.

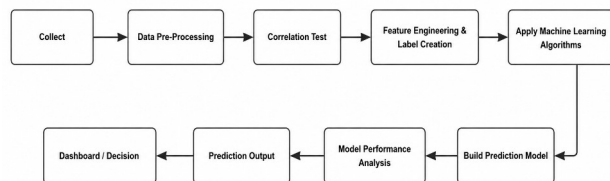


Figure 2 System Architecture

4.0 RESULTS AND DISCUSSION

Model performance was evaluated using accuracy, precision, recall, F1-score, AUC-ROC and cross-validation F1-score (see Table 3).

Table 3 Performance Comparison of Machine Learning Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC-ROC	CV F1 Mean
Random Forest	90.88	96.27	91.70	93.93	0.9651	0.9462
Gradient Boosting	90.58	96.25	91.30	93.71	0.9623	0.9437
Support Vector Machine (SVM)	89.51	97.40	88.74	92.86	0.9591	0.9358
Logistic Regression	88.91	96.76	88.54	92.47	0.9641	0.9373
Decision Tree	89.67	97.40	88.93	92.98	0.9520	0.9332
Naïve Bayes	86.02	86.57	96.84	91.42	0.8981	0.9224
K-Nearest Neighbors	88.30	95.16	89.33	92.15	0.9416	0.9237

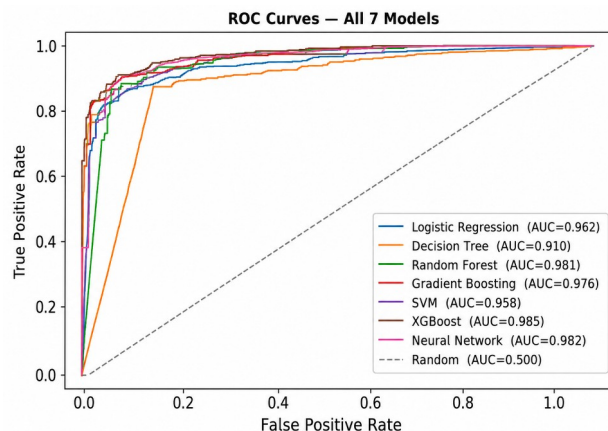


Figure 3 ROC curves for all models

As shown in Figure 3, most models demonstrate strong discrimination capability as their curves are clearly above the random baseline. Random Forest achieved the highest accuracy of 90.88% and AUC value of 0.965 indicating the best ability to distinguish between students who graduate on time and those who experience delayed graduation. Therefore, the ROC analysis confirms that Random Forest provides the most reliable predictive performance, and it was selected as the final model for hyperparameter tuning and integration into the predictive system.

Following model selection, the Random Forest model was optimized by tuning parameters including number of estimators, maximum tree depth, and minimum samples split. The optimal configuration consisted of 200 estimators with maximum depth of 10. When evaluated on the test dataset, the tuned model demonstrated high

predictive performance as it achieved a cross-validation F1-score of 94.26% which indicates strong generalization capability (see Table 4).

Table 4 Final Model Performance

Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC-ROC
91.34	96.10	92.49	94.26	0.954

Moreover, the confusion matrix in Figure 4 indicates that the model correctly classified most of the on-time and delayed graduation cases with relatively few misclassifications.

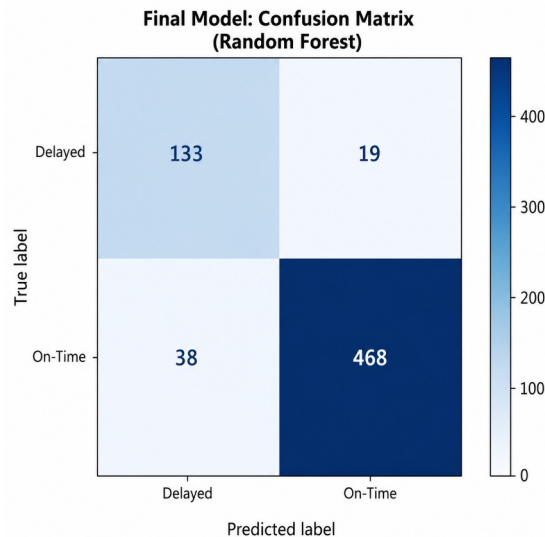


Figure 4 Confusion Matrix for Final Random Forest Model

Feature importance analysis further revealed that Academic Risk Score, CGPA, Attendance Percentage and Failed Courses are the most influential predictors in determining graduation outcomes (see Figure 5).

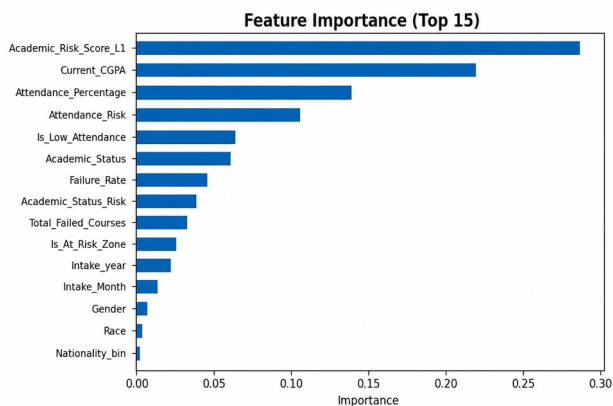


Figure 5 Feature Importance of the Random Forest Model

Finally, the trained Random Forest model was applied to 1682 currently enrolled students to generate prediction probabilities and classify them into risk tiers as in Table 5.

Table 5 Risk Tier Distribution for Current Students

Risk Category	Number of Students
High Risk	57
Medium Risk	423
Low Risk	1202
Total	1682

This risk classification provides valuable insights for academic staff and advisors. Students categorized as High Risk may require targeted academic interventions such as counselling or academic support programmes while Medium Risk students may benefit from closer monitoring to ensure they remain on track.

To support data interpretation and decision-making, the machine learning outputs were integrated into an interactive Power BI dashboard. The dashboard consists of three main sections, namely Student Overview dashboard (Figure 6), Student Performance Dashboard (Figure 7), and Graduation Risk & Intervention Dashboard (Figure 8).

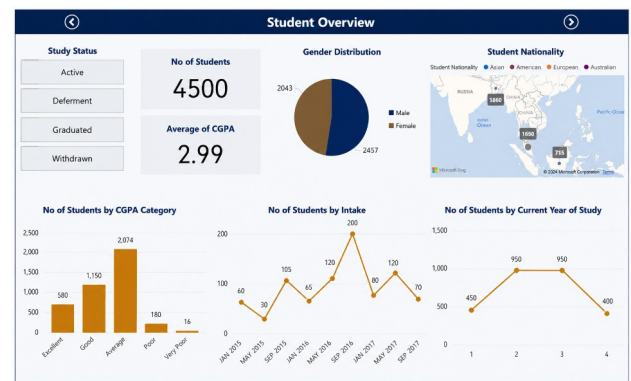


Figure 6 Student Overview Dashboard

The Student Overview dashboard provides a high-level summary of the student dataset. This dashboard focuses on demographic distribution, academic categories and enrolment trends. It allows users to quickly understand the overall composition of the student population before performing deeper performance or risk analysis.

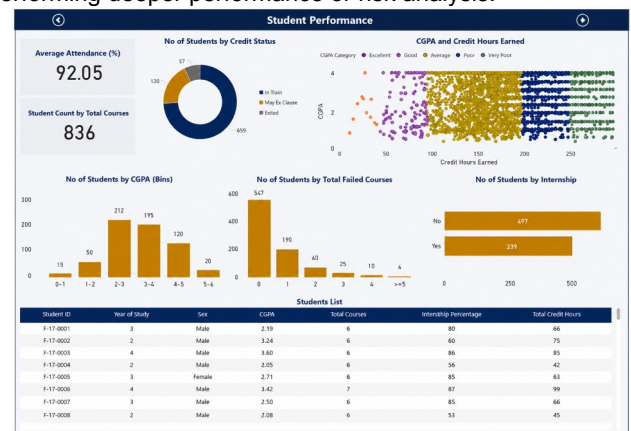


Figure 7 Student Performance Dashboard

The Student Performance dashboard provides an overview of students' academic performance and learning

behaviour. It focuses on key indicators such as attendance, course failures, credit progress, CGPA distribution and internship participation to evaluate student progression and identify potential academic challenges. A student list table is also included to display detailed academic information for each individual student.

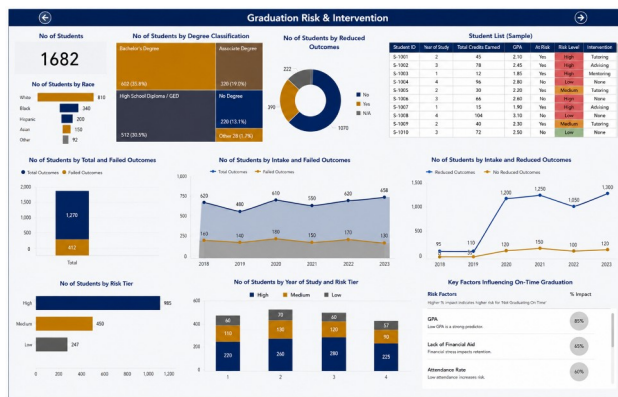


Figure 8 Graduation Risk & Intervention Dashboard

The Graduation Risk & Intervention dashboard presents the prediction results generated by the machine learning model and highlights students who may be at risk of delayed graduation. The dashboard analyses 1682 currently enrolled students and visualises their predicted graduation outcomes, academic risk levels and key influencing factors.

These dashboards enable academic staff and advisors to explore student data interactively and identify students who may require additional academic support or monitoring.

The results demonstrate that machine learning models can effectively predict student graduation outcomes based on academic progression indicators. Among the evaluated models, Random Forest achieved the best performance likely due to its ensemble learning mechanism which combines multiple decision trees to capture complex relationships between variables while reducing overfitting.

Feature importance analysis identified CGPA, course failures and attendance percentage as the most influential predictors of graduation outcomes. These variables reflect core academic performance indicators that directly affect student progression within university programmes.

Moreover, these findings are consistent with previous studies which also identified academic indicators such as GPA, course completion and attendance as key predictors of student success.

From a practical perspective, integrating machine learning predictions with Power BI dashboards enables academic staff and advisors to visualize student performance and identify at-risk students earlier. This allows institutions to implement targeted interventions and improve academic monitoring.

Overall, the study highlights the potential of predictive analytics combined with data visualization to support proactive student success management in higher education.

5.0 CONCLUSION

This study developed a machine learning-integrated Power BI dashboard to predict whether undergraduate Computer Science students will graduate on time or experience delayed graduation. Using a dataset of 4500 student records, several machine learning algorithms were implemented and evaluated with Random Forest achieving the most reliable predictive performance. The system combines predictive modelling with interactive visualization, enabling academic staff and advisors to monitor student performance and identify potential academic risks more effectively. A limitation of this study is the use of a synthetic dataset, which may not fully capture real-world variability. The proposed framework demonstrates the practical potential of integrating machine learning and business intelligence tools to support evidence-based academic management in higher education institutions.

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